

Test

Test

8/5/2021

Inverse transform method for sampling from random variable with kernel g

First, we can use the inverse-transform method. The kernel of g is $\exp(-|x|)$, and so the normalizing constant Z is given by

$$\frac{1}{Z} \int_{-\infty}^{\infty} g(x) dx = 1,$$

which may be rewritten as

$$\int_{-\infty}^0 e^x dx + \int_0^{\infty} e^{-x} dx = Z,$$

which has the solution $Z = 2$. Thus, random variable Y that has a kernel g has a density

$$f_Y(y) = \frac{1}{2} \exp(-|y|).$$

```
g.density <- function(x) { 0.5*exp(-abs(x)) }
```

The cdf F_Y is defined as

$$F_Y(y) = \frac{1}{2} \int_{-\infty}^y \exp(-|s|) ds.$$

If $y \leq 0$, then

$$F_Y(y) = \frac{1}{2} \int_{-\infty}^y \exp(s) ds = \frac{1}{2} \exp(y)$$

and if $y > 0$, then

$$F_Y(y) = F_Y(0) + \frac{1}{2} \int_0^y \exp(-s) ds = \frac{1}{2} + \frac{1}{2}(1 - \exp(-y)),$$

which may be written as the piece-wise function

$$F_Y(y) = \begin{cases} \frac{1}{2} \exp(y) & y \leq 0 \\ 1 - \frac{1}{2} \exp(-y) & y > 0. \end{cases}$$

Now, we find the inverse of F_Y . Over $y \leq 0$, the inverse of F_Y is given by

$$p = \frac{1}{2} \exp(y).$$

Solving for y , we obtain $y = \log(2p)$. Note that given $y \leq 0$, $p \in (0, 0.5]$. Over $y > 0$, the inverse of F_Y is given by

$$p = 1 - \frac{1}{2} \exp(-y).$$

Solving for y , we obtain

$$y = -\log(2 - 2p)$$

where $p \in (0.5, 1)$. We thus have a piecewise function

$$F^{-1}(p) = \begin{cases} \log(2p) & p \in (0, 0.5] \\ -\log(2 - 2p) & p \in (0.5, 1). \end{cases}$$

Thus, to sample from Y , we observe u from $U(0, 1)$ and, if $u \leq 0.5$, we let $y = \log(2u)$ and otherwise let $y = -\log(2 - 2u)$.

```
ry <- function(n)
{
  ys <- numeric(n)
  for (i in 1:n)
  {
    u <- runif(1)
    if (u <= 0.5) { ys[i] <- log(2*u) }
    else { ys[i] <- -log(2-2*u)}
  }
  ys
}
```